

MATH 199/249: TOPOLOGY I EXAM 1

FALL 2026

You have 80 minutes to do these problems in any order. There are 36 points total. Please show work unless otherwise specified.

You are allowed to look at any notes on paper that you wrote prior to the exam. However, you are not allowed to use electronic devices of any kind—including phones, computers, tablets, or audio devices—or any software, including AI tools. If you need to use the bathroom, please give any phones or other electronics to the proctor first.

Problem 1 (6 points). Let d_1 and d_2 be metrics on a set X . Decide whether or not $d(x, y) := d_1(x, y) + d_2(x, y)$ must also be a metric on X . If so, prove it; if not, give a counterexample.

Solution. It is indeed a metric.

- (a) Since $d_1(x, y), d_2(x, y) \geq 0$ for all $x, y \in X$, we have $d_1(x, y) + d_2(x, y) \geq 0$, with equality if and only if $d_1(x, y) = 0$ and $d_2(x, y) = 0$. The latter occur if and only if $x = y$. Therefore, $d(x, y) \geq 0$, with equality if and only if $x = y$.
- (b) For all $x, y \in X$, we have

$$d(x, y) = d_1(x, y) + d_2(x, y) = d_1(y, x) + d_2(y, x) = d(y, x).$$

- (c) For all $x, y, z \in X$, we have

$$\begin{aligned} d(x, z) &= d_1(x, z) + d_2(x, z) \\ &\leq d_1(x, y) + d_1(y, z) + d_2(x, y) + d_2(y, z) \\ &= d_1(x, y) + d_2(x, y) + d_1(y, z) + d_2(y, z) \\ &= d(x, y) + d(y, z). \end{aligned} \quad \square$$

Problem 2 (7 points). Let $(\alpha_n)_n$ be a sequence of positive real numbers such that $\alpha_n > n$ for all n . Do at least one of the following:

- (a) Show that $(\alpha_n)_n$ diverges in the Euclidean metric.
- (b) Show that $(1/\alpha_n)_n$ converges to 0 in the Euclidean metric.

Solution. (a) We must show that the sequence cannot converge to any fixed α in the Euclidean metric. That is: For some $\epsilon > 0$, and for all indices N , there is some $n \geq N$ such that $|\alpha - \alpha_n| \geq \epsilon$. Take $\epsilon = 1$. By the archimedean principle, there is some integer $m > \alpha$. For all N , take $n \geq \max\{N, m + 1\}$. Then $n \geq N$, and at the same time, $\alpha_n > n \geq m + 1 > \alpha$, from which

$$|\alpha - \alpha_n| \geq |\alpha - (m + 1)| \geq 1 + |\alpha - m| \geq 1,$$

as needed.

- (b) We must show that for all $\epsilon > 0$, there is some N such that for all $n \geq N$, we have $|0 - \alpha_n| < \epsilon$. Indeed, by the archimedean principle, we can pick an integer $N > 1/\epsilon$. Then for all $n \geq N$, we have $\alpha_n > n > 1/\epsilon$, from which $0 < 1/\alpha_n < \epsilon$, from which $|\alpha_n| < \epsilon$, as needed. $\square$

Problem 3 (8 points). Let e be the Euclidean metric on $\mathbf{R}^2$. Let t be the taxicab metric on $\mathbf{R}^2$ defined by

$$t(\vec{u}, \vec{v}) = |v_1 - u_1| + |v_2 - u_2|.$$

Write down radii $\delta_1, \delta_2 > 0$ such that $B_e(\vec{0}, \delta_1) \subseteq B_t(\vec{0}, 1) \subseteq B_e(\vec{0}, \delta_2)$, where $\vec{0} = (0, 0)$. Sketch these open balls, and label a few points. *Hint:* Do the t -ball first.

Solution. [Picture(s) omitted.] The t -ball is the open region bounded by the square in $\mathbf{R}^2$ with vertices at $(\pm 1, 0), (0, \pm 1)$. In the Euclidean metric, the points of the boundary closest to $\vec{0}$ occur at $(\frac{1}{2}, \pm \frac{1}{2}), (-\frac{1}{2}, \pm \frac{1}{2})$, where the Euclidean distance to $\vec{0}$ is $\frac{\sqrt{2}}{2}$, while the points of the boundary furthest from $\vec{0}$ are the vertices themselves. So any choices with

$$\delta_1 \leq \frac{\sqrt{2}}{2} \quad \text{and} \quad \delta_2 \geq 1$$

work. [In the sketch, I would at least label the vertices of the t -ball.] $\square$

Problem 4 (6 points). Below, *continuous* means either *sequentially continuous* or *analytically continuous*. Decide which of the following are true or false. If false, give a counterexample, without proof.

- (a) Every function from Euclidean $\mathbf{R}$ to Euclidean $\mathbf{R}$ is continuous.
- (b) Every constant function from Euclidean $\mathbf{R}$ to Euclidean $\mathbf{R}$ is continuous.
- (c) Every function from Euclidean $\mathbf{R}$ to discrete $\mathbf{R}$ is continuous.

Solution. (a) False. Lots of counterexamples: *e.g.*, any function that has a constant value outside 0, where it takes a different value.

(b) True.

(c) False. We showed in class that $f(x) = x$ is not sequentially continuous from Euclidean $\mathbf{R}$ to discrete $\mathbf{R}$. $\square$

Problem 5 (5 points). Let $f: \mathbf{R} \rightarrow \mathbf{R}$ be the squaring function $f(x) = x^2$.

- (a) Give an open interval (a, b) such that $f^{-1}((a, b))$ is not an open interval.
- (b) If $U \subseteq \mathbf{R}$ is analytically open, must $f^{-1}(U)$ be analytically open? (Yes/No)

Solution. (a) Take any interval (a, b) with $a > 0$. Then

$$f^{-1}((a, b)) = (-\sqrt{b}, -\sqrt{a}) \cup (\sqrt{a}, \sqrt{b}),$$

but this isn't an open interval because it omits 0, which is strictly between $-\sqrt{b}$ and $\sqrt{b}$.

(b) Yes. [We showed this in class.] □

Problem 6 (4 points). Give $\mathbf{R}^2$ the Euclidean metric. Write down at least one of the following, without proof:

- (a) An analytically open set in $\mathbf{R}^2$ that is not an open ball.
- (b) A sequence that stays within an open ball of $\mathbf{R}^2$, yet still diverges.

Solution. (a) Take any union of disjoint open balls: say, $B(0, 1) \cup B(2, 1)$. [There are also examples formed from two open balls that are not disjoint, but merely do not contain one another.]

(b) Take any oscillating sequence, like $(\alpha_n)_n$ where $\alpha_n = ((-1)^n, 0)$. [This stays within the open ball $B(0, 2)$.] □